MILITARY SPECIFICATION

TEST SET, TS-3389, OMEGA RECEIVING SET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Purpose. This specification prescribes the requirements for the design, construction, and testing of the TS-8839 OMEGA receiving set test set for use with OMEGA receiving sets AN/SRN-17 and AN/BRN-7.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore, General Specification
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment
MIL-STD-415	- Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-480	- Configuration Control-Engineering Changes, Deviations and Waivers

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MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production a sample shall be tested as specified in 4.3.

3.2 General specification requirements. The equipment shall meet the requirements of MIL-E-16400 except as otherwise specified herein.

3.3 Parts and materials. In the selection of parts and materials, the fulfillment of design objectives shall be the prime consideration. Microelectronic technology shall be used.

3.3.1 Nonstandard parts and materials. Approval for the use of nonstandard parts and materials shall be obtained in accordance with MIL-E-16400 and as specified herein. Throw-away modules shall be considered parts, and nonstandard part approval shall be obtained for each.

3.3.2 Modular construction of equipment. The equipment shall be modularized in accordance with MIL-E-16400 and as specified herein.

3.3.3 Mercury contamination. Components shall not have come in contact with mercury, mercury compounds, or mercury-bearing instruments or devices, employing only a single boundary of containment during manufacture, tests or inspections without the specific written approval of the command or agency concerned.

3.3.4 Plastic encapsulated devices. Plastic encapsulated devices may be used in throw-away assemblies with the written approval of the command or agency concerned. Conformal coating of throw-away assemblies using plastic encapsulated devices shall be required. Conformal coatings shall be in accordance with MIL-E-16400.

3.4 Design and construction. The TS-3389 equipment shall be completely self-contained, including interconnecting cabling and connectors, and built into a suitable storage/transportation container.

3.4.1 Configuration control. Configuration control shall be maintained in accordance with MIL-STD-480.

3.5 Reliability. The TS-3389 shall have a mean-time-between-failure (MTBF) of 1000 hours (θ_o) in accordance with MIL-STD-781.

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3.6 Maintainability. The receiving set specified herein shall have an equipment repair time (ERT) of no greater than 30 minutes when repair is accomplished by replacement of modules and parts at the organizational level. The ERT includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all module and part repairs.

3.7 Environmental conditions. Environmental conditions shall be in accordance with MIL-E-16400 and as specified herein.

3.7.1 Temperature classification. Temperature classification of the equipment shall be range 4. The lowest operating temperature will be 5° C.

3.7.2 Wind speed, icing, hydrostatic pressure, hydraulic shock and nuclear air blast. The wind speed, icing, hydrostatic pressure, hydraulic shock and nuclear air blast requirements of MIL-E-16400 are not applicable to the TS-3389.

3.7.3 Electromagnetic interference. Electromagnetic interference (EMI) requirements shall be in accordance with MIL-STD-461, and as specified herein. The equipment shall be considered class IC.

3.7.4 Shock. Shock requirements are in accordance with MIL-S-901 and as specified herein. The equipment is classified as deck mounted. A combination of Grade A and Grade B shockproofness is required as follows:

(a) The Grade A shockproofness requirement of the equipment is limited to one foot hammer blow.
(b) The Grade B shockproofness requirement of the equipment applies to the equipment when its enclosure is secured.

3.7.5 Vibration. Vibration requirements shall be in accordance with MIL-STD-167 and as specified herein. With the enclosure of the equipment secured, the equipment shall meet the type I requirements up to and including 33 hertz (Hz).

3.7.6 Enclosure. Enclosure requirements shall be in accordance with MIL-STD-108 and as specified herein. With the enclosure of the equipment secured, the degree of enclosure shall be watertight.

3.8 Electrical design. Electrical design shall be in accordance with MIL-E-16400 and as specified herein.

3.8.1 Power. Power will be provided directly from the ON-128 interconnecting group of the AN/BRN-7 receiving set. Primary power will be 115 volts (V), 400 Hz, single phase.

3.8.2 Interface requirements. The TS-3389 shall interface directly with the AN/BRN-7 (or AN/SRN-17) receiving set. Cabling and connectors (which mate with those provided on the AN/BRN-7 receiving set) shall be provided as part of the TS-3389.

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3.8.2.1 Power interface. A single three conductor cable shall be used to provide the power interface between the TS-3389 and the interface box. A quick disconnect connector shall be used.

3.8.2.2 Functional interface. The functional interface shall consist of a two cable interface between the TS-3389 and the OR-133 receiver computer (of the AN/BRN-7 receiving set). One male and one female quick disconnect type connector shall be used. All control and data transfer capabilities necessary to meet the performance requirements specified herein shall be included in this interface.

3.9 Mechanical design. The mechanical design shall be in accordance with MIL-E-16400 and as specified herein. The equipment shall be designed into a suitcase type enclosure suitable for storage and transportation. All interconnecting wiring, including that for the interface with the OR-133 and all program tapes necessary to meet the performance requirements specified herein, shall be stored within the enclosure. The interconnecting cables for interface with the OR-133 shall be 5 feet (150 cm) in length. Except as otherwise noted, the following mechanical design requirements apply to the equipment in the nonoperating mode and secured for storage.

3.9.1 Size and weight. Total weight shall not exceed 40 pounds. Total size shall not exceed 12 inches (304.8 mm) by 18 inches (457.2 mm).

3.9.2 Maintenance design. The TS-3389 shall be designed to permit accomplishment of all corrective maintenance at the organizational level. Organizational level maintenance will consist of replacing modular assemblies and chassis mounted components by shipboard technicians. No piece part or printed circuit board repairs will be done at the organizational level. No special test equipment shall be required for maintenance.

3.9.3 Circuit test points. Circuit test point accesses shall be provided in accordance with MIL-STD-415. Class A and class B test provisions shall be provided on depot repairable modules (if any).

3.10 Controls, indicators and panel layout. Controls, indicators and panel layout shall be in accordance with MIL-E-16400 and as specified herein.

3.10.1 Panel layout. Controls and indicators shall be kept to a minimum consistent with the performance requirements as specified herein.

3.10.2 Indicators. Lamps and other indicators shall be chosen in accordance with MIL-STD-1472. Indicators shall have opaque backgrounds with illuminated red legends.

3.11 Performance characteristics. Performance characteristics shall be based on the intended use (see 6.1) and performance requirements specified herein. Performance requirements apply to the TS-3389 when it is interfaced with the AN/BRN-7 or AN/SRN-17 OMEGA receiving sets.

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3.11.1 Programming. The TS-3389 shall be capable of loading the computer programs. The computer programs shall be stored on magnetic tape cassettes (to be supplied with the TS-3389).

3.11.2 Fault isolation. The TS-3389 shall control the diagnostic functions of the AN/BRN-7 and AN/SRN-17. It shall be capable of confirming built-in test malfunction indications (in the receiving set) and isolating the fault to a replaceable assembly within the OR-133.

3.11.3 Performance check. The TS-3389 shall be capable of verifying that the receiving set is operating correctly. It will be used periodically for performance checks. When used in this mode, the TS-3389 shall be at least 98 percent effective.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms specified herein shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The methods of examination and testing shall be classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see 4.4.2)

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(3) Environmental inspection (see 4.4.3)
(4) Inspection for preparation for delivery (see 4.4.4)

(c) Reliability testing (see 4.5)

4.3 First article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the examination and testing specified in production inspection, production control inspection, and environmental inspection.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the requirements of Table I.

Table I. Production inspection.

Inspection	Reference
Programming	3.11.1
Performance checks	3.11.3

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to Table II below and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table II. Production control inspection.

Inspection	Reference	AQL Percent
Size and weight	3.9.1	6.5
Supply line voltage and frequency	3.8.1	6.5
Heat test	MIL-E-16400	6.5
Fault isolation	3.11.2	6.5

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4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lots using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to, and passed production control inspections may be accepted on the contract provided they are resubjected to, and pass production inspection after repair of all visible damage.

4.4.3 Environmental inspection. Environmental inspection shall be as listed in Table III and shall be performed on units which have been subjected to and passed production inspection.

Table III. Environmental inspection.

Inspection	Reference
Temperature	3.7.1
Humidity	3.7
Electromagnetic interference	3.7.3
Shock	3.7.4
Vibration	3.7.5
Inclination	3.7
Structural integrity	3.1
Salt spray	3.7
Enclosure	3.7.6

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments produced or fraction thereof. The first sample equipment shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the command or agency concerned the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth.

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed environmental tests may be accepted on the contract provided they are resubjected to, and pass production inspection after repair of all damage.

4.4.4 Inspection for preparation for delivery. Inspections shall be conducted to insure conformance with the requirements of section 5 of this specification.

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4.5 Reliability testing. The reliability testing of the TS-3389 shall be performed in accordance with MIL-STD-781.

4.5.1 Failure. All failures will be considered relevant unless determined by the command or agency concerned to be non-relevant or caused by a condition external to the equipment under test in accordance with the failure categories paragraph of MIL-STD-781 (see 6.3).

4.5.2 Reliability production acceptance (sampling) phase. Reliability sampling testing shall be conducted throughout production to insure continued production of acceptable equipment. Testing shall be in accordance with test plan XXIX of MIL-STD-781. Testing shall be at test level B except the vibration frequency shall be between 20 Hz and 33 Hz. If the reliability production acceptance phase results in a reject situation, corrective action shall be taken in accordance with MIL-STD-781. There shall be no on/off cycling.

4.5.3 Final restoration and disposition of tested equipment. Upon completion of reliability testing, the tested equipment shall be returned to production for reinspection and reconditioning. The sample may be delivered as part of the contract quantity if it meets all terms and conditions of the contract for acceptance.

4.6 Special test procedures. The following special test procedure requirements apply to the production, production control and environmental inspections. The inspection category to which the special test procedure requirement applies is listed in the inspection table as a special test reference.

4.6.1 General test requirement. The test procedure shall be in accordance with the applicable requirements of MIL-E-16400.

4.6.2 Panel layout. Panel layout shall be inspected to insure conformance with the approved panel layout drawings.

4.6.3 OMEGA receiving set. Either an AN/SRN-17 or AN/BRN-7 may be used to determine acceptable performance during the programming, fault isolation and performance check tests.

4.6.4 Fault isolation and performance check tests. A sample of 10 faults (selected randomly) shall be introduced into the AN/SRN-17 or AN/BRN-7 receiver. Using the technical manual as a guide, the faults shall be isolated using the programmer-controller. After removal of the fault, a performance check shall be made using the technical manual as a guide.

4.6.5 Vibration test. The following clarifications of the vibration test requirements are applicable:

(a) Subsequent to the test, the equipment shall meet the performance requirements specified herein.

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(b) The equipment shall be strapped to the vibration table with its enclosure secured for storage.
(c) The equipment shall be de-energized during the test.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by the suppliers shall be specified in the individual order.)

5.1 Preservation and packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. A TS-3389 programmer-controller is available to each AN/SRN-17 and AN/BRN-7 installation. The programmer-controller is used by the maintenance technician to:

(a) Load the computer program into the receiver-computer.
(b) Isolate faults to a replaceable module of the receiver-computer.
(c) Verify performance of the receiver-computer is within specified tolerances.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Marking, levels of preservation, and packaging and packing required (see 5.1).

6.3 Definitions.

6.3.1 Failure. Any malfunction or deviation which prevents the equipment from performing in accordance with the performance requirements set forth in this specification.

6.4 Supersession data. This specification includes the requirements of Naval Electronic Systems Command contract specifications ELEX-T-228 and ELEX-P-128.

Preparing activity:
Navy - EC

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